

Marketing Funnel & Conversion Performance Analysis

Statistical Analysis with Business Recommendations

Future Interns

Data Science & Analytics - Task 3
Marketing Funnel & Conversion Analysis

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1 Executive Summary

This report analyzes **41,188 marketing contacts** from a bank's telemarketing campaigns to identify conversion drivers, funnel drop-off points, and optimization opportunities. Using logistic regression, we quantify the strongest predictors of customer subscription and provide evidence-based recommendations to improve marketing ROI.

1.1 Funnel Overview

Metric	Value
Top of Funnel (Contacts)	41,188
Bottom of Funnel (Conversions)	4,640
Conversion Rate	11.27%
Drop-off Rate	88.73%

Table 1: Marketing Funnel Overview

Critical Finding: 88.73% of contacts do not convert. The #1 lever for improvement is **call duration** — calls over 15 minutes convert at 59.4% vs 1.28% for calls under 2 minutes. Other key drivers include contact method (cellular 3× better than telephone), seasonality (March 8× better than May), and previous relationship (65% re-conversion).

2 Data Overview

Attribute	Description
Dataset Source	Bank Marketing Campaign Dataset
Total Contacts	41,188
Features	21 (demographics, campaign, economic)
Target Variable	Subscription (yes/no)
Time Period	May 2008 - November 2010
Data Quality	12 duplicates, unknown values in job/education

Table 2: Dataset Characteristics

2.1 Business Questions Addressed

Q1: Where are contacts dropping off in the funnel?

Q2: Which channels bring high-quality leads?

Q3: How can conversion rates be improved?

Q4: Which segments should be prioritized?

Q5: What is the ROI impact of optimization?

3 Funnel Analysis Results

3.1 Conversion by Contact Method

Contact Method	Contacts	Conversions	Rate
Cellular	26,144	3,853	14.74%
Telephone	15,044	787	5.23%

Table 3: Conversion Rate by Contact Method

Key Finding: Cellular is 3× More Effective

Switching from telephone to cellular contact improves conversion odds by **66%**. The business should prioritize cellular outreach and gradually phase out telephone-only campaigns.

3.2 Conversion by Job Segment

Job	Contacts	Conversions	Rate
Student	875	275	31.43%
Retired	1,720	434	25.23%
Unemployed	1,014	144	14.20%
Admin	10,422	1,352	12.97%
Blue-Collar	9,254	638	6.89%

Table 4: Conversion Rate by Job Segment

3.3 Conversion by Age Group

Age Group	Contacts	Conversions	Rate
60+	1,193	472	39.56%
Under 30	5,669	922	16.26%
50-59	6,862	697	10.16%
30-39	16,938	1,715	10.13%
40-49	10,526	834	7.92%

Table 5: Conversion Rate by Age Group

3.4 Conversion by Call Duration

Call Duration	Contacts	Conversions	Rate
15+ minutes	1,284	762	59.35%
10-15 minutes	2,194	928	42.30%
5-10 minutes	7,772	1,441	18.54%
2-5 minutes	17,143	1,345	7.85%
0-2 minutes	12,795	164	1.28%

Table 6: Conversion Rate by Call Duration

Critical: Call Duration is the #1 Predictor

Calls over 15 minutes convert at **46×** the rate of calls under 2 minutes (59.4% vs 1.28%). Every additional minute on the phone dramatically improves conversion probability. Training agents to extend quality conversations represents the highest-ROI intervention.

3.5 Conversion by Month (Seasonality)

Month	Contacts	Conversions	Rate
March	546	276	50.55%
December	182	89	48.90%
September	570	256	44.91%
October	718	315	43.87%
May	13,769	886	6.43%

Table 7: Conversion Rate by Month

3.6 Conversion by Previous Outcome

Previous Outcome	Contacts	Conversions	Rate
Success	1,373	894	65.11%
Failure	4,252	605	14.23%
Nonexistent	35,563	3,141	8.83%

Table 8: Conversion Rate by Previous Campaign Outcome

4 Statistical Analysis

4.1 Logistic Regression Model

Model: Conversion $\sim$ Contact + Job + Age + Duration + Month + Previous Outcome + Education

Metric	Value
Pseudo R-squared	0.371
Accuracy	90.60%
Specificity	97.36%
Sensitivity	37.41%
AIC	18,324

Table 9: Logistic Regression Model Performance

4.2 Top Conversion Drivers (Odds Ratios)

4.2.1 Factors That INCREASE Conversion

Factor	Odds Ratio	Interpretation
Call 15+ minutes	228×	Massive conversion boost
Call 10-15 minutes	97.4×	Huge positive effect
Call 5-10 minutes	21.8×	Strong driver
Previous Success	10.7×	Retargeting gold
Month: March	6.15×	Seasonal peak
Age 60+	3.31×	Senior advantage
Month: October	2.90×	Fall opportunity

Table 10: Top Conversion Increaseers

4.2.2 Factors That DECREASE Conversion

Factor	Odds Ratio	Interpretation
Telephone (vs Cellular)	0.34×	66% lower conversion
Month: July	0.38×	62% lower conversion
Month: May	0.43×	57% lower conversion
Month: November	0.46×	54% lower conversion
Job: Blue-Collar	0.68×	32% lower conversion

Table 11: Top Conversion Reducers

Statistical Conclusion

Call duration is overwhelmingly the strongest predictor of conversion — **228×** odds ratio for 15+ minute calls. Combined with cellular contact (3× better), seasonal timing (March 6× better), and retargeting previous successes (10.7× better), the model identifies clear levers for improving the 11.27% baseline conversion rate.

5 Recommendations

5.1 Immediate Actions (0–3 Months)

1. **Prioritize cellular contact** — 3× higher conversion; phase out telephone-only outreach

2. **Train agents on call duration** — Extend conversations past 5 minutes (18.5%+ conversion threshold)
3. **Retarget previous successes** — 65.1% re-conversion rate is the highest-ROI segment

5.2 Strategic Initiatives (3–12 Months)

4. **Target students and retirees** — 31.4% and 25.2% conversion rates respectively
5. **Optimize seasonal campaigns** — Shift volume to March, October, December (40-50% conversion)
6. **Reduce May/July campaigns** — 6.4% and 9.1% conversion rates waste resources
7. **Develop 60+ age segment** — 39.6% conversion rate; create specialized messaging

5.3 Estimated Impact

Improving the baseline conversion rate from **11.27% to 15%** (a 33% improvement) through these interventions would generate approximately **1,535 additional conversions** from the same contact volume. At an estimated customer lifetime value of \$500, this represents **\$767,500 in additional revenue**.

6 Methodology

Technique	Purpose
Data Cleaning	Handle missing values, create derived features
Descriptive Analysis	Conversion rates by segment, KPI calculation
Funnel Analysis	Identify drop-off points across segments
Logistic Regression	Quantify conversion drivers (odds ratios)
Confusion Matrix	Evaluate model performance
Seasonality Analysis	Identify optimal campaign timing

Table 12: Analytical Methodology

Tools Used: R Statistical Computing Environment, tidyverse, broom, ggplot2

7 Conclusion

This analysis identified an **11.27% baseline conversion rate** from 41,188 marketing contacts. Statistical modeling revealed clear, actionable levers for improvement.

The Conversion Formula: The highest-probability contact uses **cellular phone**, has a **previous successful relationship**, is called for **15+ minutes**, during **March or October**, and is a **student or retiree aged 60+**. This profile converts at rates exceeding 50%.

Key Takeaways:

- **Call duration** is the #1 predictor — 228× odds ratio for 15+ minute calls
- **Cellular crushes telephone** — 3× better conversion (14.7% vs 5.2%)
- **Retargeting works** — Previous successes convert at 65.1%
- **Timing matters** — March (50.6%) is 8× better than May (6.4%)
- **Students and seniors** are gold — 31.4% and 39.6% conversion respectively

The recommendations provide a clear roadmap for improving marketing ROI. By focusing on the right contact method, extending call duration, targeting high-conversion segments, and optimizing seasonal timing, the business can realistically improve conversion rates by 33%+.

Estimated Revenue Impact: +\$767,500 from conversion optimization

Report prepared for **Future Interns - Data Science & Analytics Task 3**

Analysis conducted using R Statistical Computing Environment

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